Discuss and recall — the Grade 5 course

An entry diagnostic: what Grade 5 left behind, and what Grade 6 will need on the first day.

LESSON 1 | 1 × 40 MIN | MATEMATIKA 6, KIRISH | STAGE 6 DIAGNOSTIC

LESSON CLOCK

6' | 20' | 10' | 4'

What Grade 5 covered	6 min
The diagnostic	20 min
Answers and diagnosis	10 min
The year ahead	4 min

LEARNING OBJECTIVES

- Recall the four operations on whole numbers and on fractions.
- Recall the order of operations and the use of brackets.
- Recall perimeter, area and the units used for each.
- Identify, from the diagnostic, what needs revising this month.

TEXTBOOK

National	pp. 3–8
Cambridge	Stage 7, pp. 1–6
Workbook	Entry test

01 Explanation

What Grade 5 covered

BLOCK	WHAT YOU SHOULD BE ABLE TO DO
whole numbers	add, subtract, multiply and divide in columns
order of operations	brackets, then × and ÷, then + and −
fractions	simplify, compare, add and subtract with a common denominator
decimals	read, order, add and subtract
measurement	perimeter and area of rectangles, and the units
angles	measure with a protractor, and name acute, right and obtuse

NOTHING HERE IS FINISHED

Every one of these blocks returns this year in a longer form: fractions are divided, decimals are multiplied, and area moves from rectangles to circles.

The diagnostic — 20 marks, 20 minutes

Q	TASK	MARKS
1	$348 + 2\,976$ and $5\,004 - 1\,897$	*2*
2	$47 \cdot 26$ and $918 \div 27$	*3*
3	$12 + 3 \cdot (8 - 5)$	*2*
4	Simplify $\frac{18}{24}$ and compare $\frac{3}{5}$ with $\frac{5}{8}$	*3*
5	$\frac{2}{3} + \frac{1}{4}$	*3*
6	$4.7 + 2.85$ and $9.1 - 3.62$	*3*
7	A rectangle $7\ cm$ by $4\ cm$: its perimeter and area	*4*

THE ANSWERS

$3\,324$; $3\,107$; $1\,222$; 34; 21; $\frac{3}{4}$ and $\frac{3}{5} < \frac{5}{8}$; $\frac{11}{12}$; 7.55 and 5.48; $22\ cm$ and $28\ cm^2$.

Diagnosis

IF Q… WAS WRONG	REVISE	BEFORE
1–2	column arithmetic	the first week
3	the order of operations	algebraic expressions, lesson 8
4–5	simplifying and adding fractions	dividing fractions, lesson 22
6	decimals	the Cambridge decimal block, lesson 76
7	perimeter and area	the circle, lesson 92

A GAP LEFT OPEN IN SEPTEMBER CLOSES NOTHING BY ITSELF

Each row above names the lesson where the missing skill is needed again. That lesson is the deadline — not the end of the year.

The year ahead

QUARTER	MAIN BLOCKS	HOURS
I	directed numbers, expressions and equations, dividing fractions, ratio	*54*
II	percentages, decimals, angles, the circumference of a circle	*42*
III	the area of a circle, compound figures, speed, volume	*60*
IV	data and pie charts, probability, solids and nets, revision	*48*

SIX HOURS A WEEK, NOT FIVE

This class has one lesson a week more than the national plan needs. That hour carries Cambridge Stage 7 work — negative numbers, decimals, data and probability — which the national Grade 6 programme does not cover at all.

02 Worked examples

EXAMPLE 1 **Compute $12 + 3 \cdot (8 - 5)$.**

1. Brackets first: $8 - 5 = 3$.
2. Then multiply: $3 \cdot 3 = 9$.
3. Then add: $12 + 9 = 21$.

ANSWER

21

EXAMPLE 2 **Compute $\frac{2}{3} + \frac{1}{4}$.**

1. The lowest common denominator is *12*.

2. $\frac{8}{12} + \frac{3}{12}$

3. $= \frac{11}{12}$

ANSWER

$\frac{11}{12}$

EXAMPLE 3 **A rectangle is *7 cm* by *4 cm*. Find its perimeter and area.**

1. $P = 2(7 + 4) = 22$ cm

   A length.

2. $S = 7 \cdot 4 = 28$ cm^2

   An area — square units.

ANSWER

22 cm and *28 cm*^{2}

03 Interactive model

Mark the diagnostic in the lesson and hand back the diagnosis table rather than a score; a number tells a pupil nothing they can act on.

What Grade 5 left behind

$12 + 3 \cdot (8 - 5)$ equals:

- 45
- 21
- 27
- 15

$\frac{18}{24}$ simplifies to:

- $\frac{3}{4}$
- $\frac{9}{12}$
- $\frac{2}{3}$
- $\frac{6}{8}$

$\frac{2}{3} + \frac{1}{4}$ equals:

- $\frac{3}{7}$
- $\frac{11}{12}$
- $\frac{2}{12}$
- $\frac{3}{4}$

Which is larger, $\frac{3}{5}$ or $\frac{5}{8}$?

- $\frac{3}{5}$
- $\frac{5}{8}$
- equal
- cannot say

$4.7 + 2.85$ equals:

- 7.55
- 6.55
- 7.12
- 4.98

A 7×4 rectangle has perimeter:

- 11
- 22
- 28
- 14

And area:

- 11
- 22
- 28
- 44

Area is measured in:

- cm
- cm^2
- cm^3
- no units

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Natural number	Natural son	Натуральное число
Fraction	Kasr	Дробь
Numerator	Surat	Числитель
Denominator	Maxraj	Знаменатель
Order of operations	Amallar tartibi	Порядок действий
Perimeter	Perimetr	Периметр
Area	Yuza	Площадь
Diagnostic	Tashxis ishi	Диагностическая работа

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The order of operations puts first:

- addition
- brackets
- multiplication
- subtraction

$\frac{18}{24}$ simplifies to:

- $\frac{3}{4}$
- $\frac{2}{3}$
- $\frac{1}{2}$
- $\frac{4}{3}$

To add fractions you need:

- the same numerator
- a common denominator
- nothing
- a decimal

Perimeter is measured in:

- cm
- cm^2
- cm^3
- degrees

$918 \div 27$ equals:

- 32
- 34
- 36
- 27

This year has how many hours of mathematics?

- 170
- 204
- 136
- 240

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $348 + 2\,976$

ANSWER $3\,324$

2. $5\,004 - 1\,897$

ANSWER $3\,107$

3. $47 \cdot 26$

ANSWER $1\,222$

4. $918 \div 27$

ANSWER 34

5. $12 + 3 \cdot (8 - 5)$

ANSWER 21

6. $\frac{18}{24}$ simplified

ANSWER $\frac{3}{4}$

7. $4.7 + 2.85$

ANSWER 7.55

Medium · the standard question

7 PROBLEMS

1. $\frac{2}{3} + \frac{1}{4}$

ANSWER $\frac{11}{12}$

2. Compare $\frac{3}{5}$ and $\frac{5}{8}$

ANSWER $\frac{3}{5} < \frac{5}{8}$

3. $9.1 - 3.62$

ANSWER 5.48

4. A 7×4 rectangle: perimeter and area

ANSWER $22\ cm,\ 28\ cm^2$

5. $100 - 4 \cdot (7 + 8)$

ANSWER 40

6. $\frac{5}{6} - \frac{1}{3}$

ANSWER $\frac{1}{2}$

7. A square of side 9: its area

ANSWER $81\ cm^2$

Hard · stretch and reason

7 PROBLEMS

1. $(24 + 36) \div (10 - 4)$

   ANSWER 10

2. $2 \cdot 3^2 + 4$

   ANSWER 22

3. A rectangle of area 48 and width 6: its length

   ANSWER 8

4. A rectangle of perimeter 30 and width 7: its length

   ANSWER 8

5. $\frac{3}{4}$ *of 60*

   ANSWER 45

6. Order $\frac{1}{2}, \frac{2}{5}, \frac{3}{5}$

   ANSWER $\frac{2}{5} < \frac{1}{2} < \frac{3}{5}$

7. How many hours a week does this class have?

   ANSWER 6

07 Homework

Homework — 5 tasks

Rework, in full, every diagnostic question you lost a mark on.

1. Rework every diagnostic question you got wrong.
2. Compute $45 + 5 \cdot (12 - 8)$ and $(45 + 5) \cdot 12 - 8$, and say why they differ.
3. Simplify $\frac{24}{36}$ and $\frac{45}{60}$.

4. Compute $\frac{3}{4} + \frac{1}{6}$.

5. A rectangle is $9\ cm$ by $5\ cm$. Find its perimeter and area with units.